

**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
(Deemed University)

**MTech/MS/PhD EXAMINATIONS 2023**

**Course: RM & IPR**

**Duration: 03 hr**

**Credits: 02**

**Code: PGC601**

**Max Mark: 50**

**Note:**

1. Marks are indicated against each question.
2. Write clearly and legibly. All diagrams (if any) to be neatly labeled.
3. Non-programmable calculator is allowed.
4. Make necessary assumptions wherever required.
5. Answer all questions

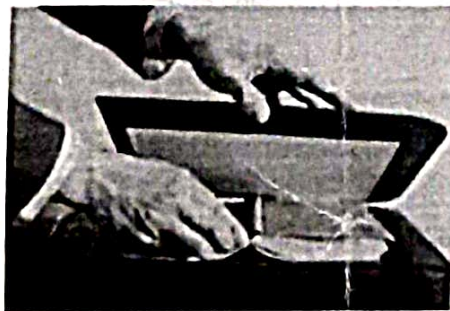
**Q1: What are the different types of research? What are the various components of research in the figure shown below?**

**5 Marks**



**Q2. What are the principles of research ethics? Comment on the figure shown below:**

**5 Marks**

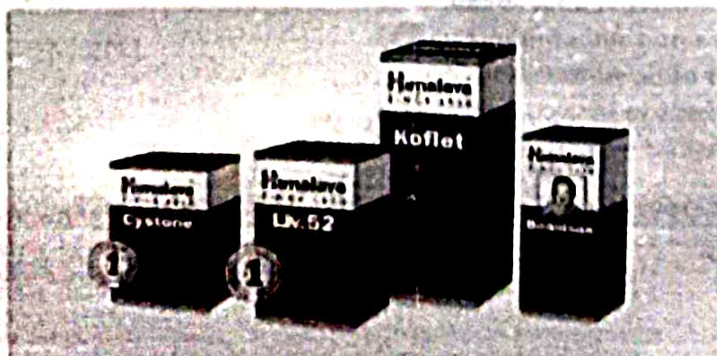


**Q3. Write short notes on TRIPS? Explain the importance of TRIPS?**

**5 Marks**

**Q4. What are the various attributes of copyrights? Explain how long will the rights exist for the image shown below:**

**5 Marks**





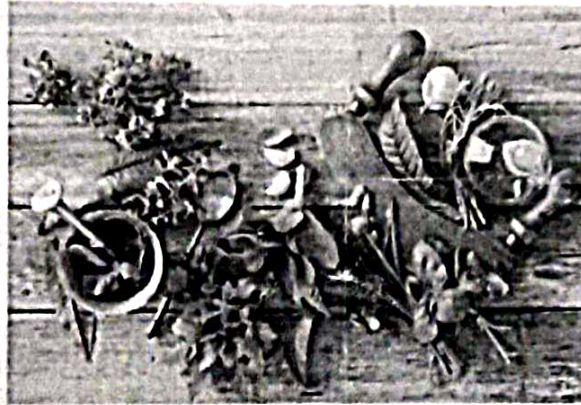
**Q5. Write short notes on hypothesis? Explain how research hypothesis supported the product development of product shown below:**

**5 Marks**



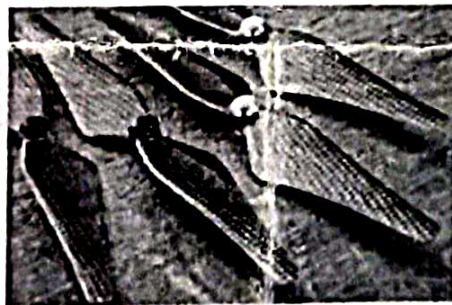
**Q6. What is the essence of research and market search? Comment on the literature shown below:**

**5 Marks**



**Q7. What are the various sources of literature? Explain the state-of-the-art search of the fibre reinforced composite for Unmanned Ariel Vehicle shown in figure?**

**5 Marks**



**Q8. What are the roles of ethical committee? What are the various conditions on patent applicants?**

**5 Marks**

**Q9. Performance analysis of mixed RF/FSO systems with STBC users "Optics Communications", Volume 381, 2016, pp. 244-252**

**Jianfeng Feng \*, Xiaohui Zhao**

**5 Marks**

- Comment on the name of journal?
- Who is the first and corresponding author?
- What is the role of second author in this paper?
- Who is the publisher of this article?
- Who is having the rights to give copyright permission?
- Who is holding the entire Intellectual properties?

**Q10. What are the different types of search engines that are used to understanding the state-of-the-art search? Write short notes on Scopus and explain how it supports in filtering documents by subject area?**

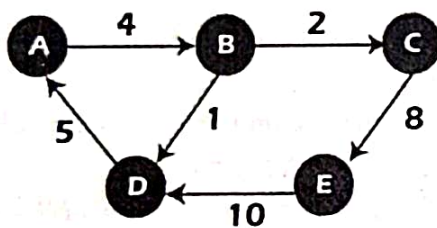
**5 Marks**

**Assessment No .1, Subject: Data Structures and Algorithms**  
**Answer all Questions, Each Question carries 2.5 marks each**

1. Explain the concept of a single linked list, and write a pseudocode for inserting the new node at the end of the single linked list.
2. Explain the concept of Queue data structure, and write a pseudocode for enqueue and dequeue operations.
3. Explain the concept of structures in C, and provide an example of how to declare and use a structure to represent information about a student, including their name, age, and grade. Additionally, describe how to access and modify the members of a structure in C
4. Explain the Stack data structure and what are the fundamental operations that can be operated on the stack.

**Assessment No .2 , Subject: Data Structures and Algorithms, Answer all Questions**

1. If the inorder traversal of a binary tree gives the elements in the order: 7,1, 6,3,5, 2, 4. (assume every node has two child nodes), then draw its tree data structure, and give the postorder traversal for the same.
2. Explain what is meant by AVL Tree. Briefly explain different operations performed on Binary search tree to make it an AVL tree with diagrams.
3. For the graph shown below, write the adjacency list and adjacency matrix representation.

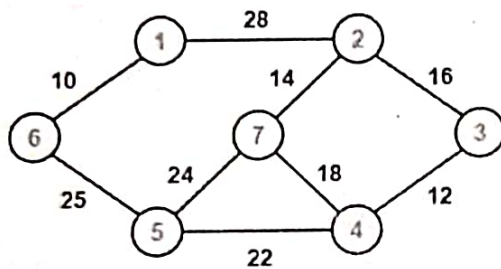


4. Explain the need for asymptotic notation for analysing the algorithms. What are the different notions used for analysing the algorithms
  5. Write a Binary search algorithm using the Iterative and recursive methods.
-

### Assessment No 3, Sub: Data Structures and Algorithms

1. Construct the minimum spanning tree for the below graph using the Prim's algorithm

3M



2. Explain what is unbounded knapsack problem. Use the greedy algorithm to find the maximum profit for a knapsack which have the capacity 16 kg for the below data

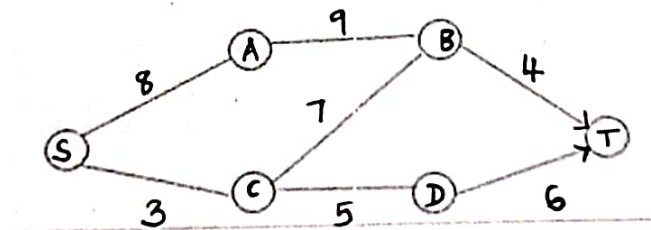
4M

Breakable

|             |      |   |   |    |      |      |   |
|-------------|------|---|---|----|------|------|---|
|             | 2.66 | 5 | 4 | 3  | 1.75 | 2.66 | 3 |
| item        | A    | B | C | D  | E    | F    | G |
| weight (kg) | 3    | 1 | 2 | 5  | 4    | 3    | 2 |
| profit      | 8    | 5 | 8 | 15 | 7    | 8    | 6 |

3. Find out the maximum flow capacity between the vertex S and T, except for S and T, every other node can send an equal amount of stuff through it.

3M





# DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University), GIRINAGAR, PUNE – 411 025

## END SEMESTER EXAMINATION: AUTUMN 2023

Course Name: Data Structures and Algorithms

Course Code: AM 608

Duration: 03 Hrs

Max Marks: 50

Note:

1. Before attempting, please ensure that you have the correct question paper.
2. Marks are indicated against each question. Calculators are only allowed.
3. Write clearly and legibly. All diagrams to be neatly labeled wherever applicable.
4. Answer all questions.

[6+2+2]

- Q1) a) Write a c Program to print the nth Fibonacci number based on the Iterative and Recursive Methods  
b) write a pseudo code for swapping of two numbers without using the temporary storage  
c) What is the output of the following code

```
#include <stdio.h>
#include <stdlib.h>
int main(){
    int n[ 10 ]={1,2,3,'d'};
    printf("%d ", n[4] );
    printf("%d ", n[5] );
    return 0;
}
```

Q2)

[ 5+ 3 +2]

- a) Explain in detail about the linear and nonlinear data structures with examples, advantages and limitations of linear and nonlinear data structure
- b) Write an algorithm for storing the values {'a', 'k', 'i', 'o', 'd'} using a doubly linked list.
- c) Write an algorithm for inserting a new data item into the singly list by considering all possible scenarios (starting, ending, and middle)

Q3)

[5+5]

- a) Explain the characteristics of the AVL tree, and give the step-by-step procedure (with diagrams) followed to implement the AVL tree for the following sequence H, I, J, B, A, E, C
- b) Explain in detail about the need for different Asymptotic Notation used in complexity analysis with advantages and limitations

Q 4)

[5 + 5]

- a) Use the counting sort algorithm, to sort the following elements with a step-by-step procedure and diagrams.

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 4 | 3 | 5 | 8 | 1 | 7 | 6 |
|---|---|---|---|---|---|---|

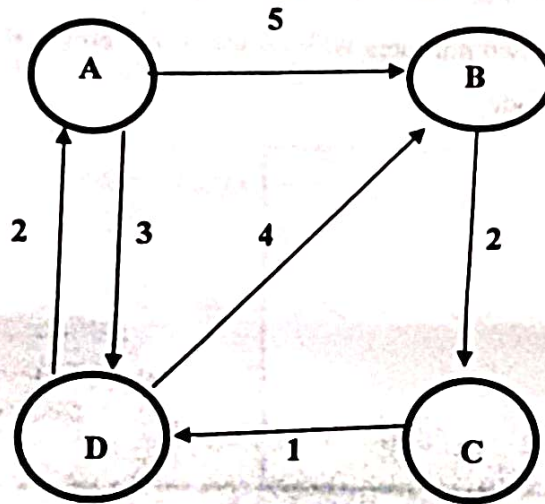
- b) Explain the working of the merge sort algorithm and report the worst-case time complexity. Sort the below array using the merge sort algorithm

|   |   |    |    |   |   |   |
|---|---|----|----|---|---|---|
| 2 | 7 | 12 | 10 | 9 | 8 | 5 |
|---|---|----|----|---|---|---|

Q 5)

[5 + 5]

- a) For the following graph find out the shortest path between all the pair of all vertices



- b) What is the maximum value that can be obtained for the unbreakable objects for a knapsack capable of carrying a weight of 6 kg using the dynamic programming and list the what are those items

| Item        | Laptop | Guitar | Phone | Printer |
|-------------|--------|--------|-------|---------|
| weight (kg) | 1      | 2      | 1     | 3       |
| Value (k\$) | 2      | 1      | 3     | 0.5     |

-----END OF THE QUESTION PAPER-----



**Assessment No .2, Subject: Data Analysis and Visualization**

1. Explain in detail about the curse of dimensionality.
2. Explain briefly about different methods that can employed to reduce the dimensionality of the features
- 3.

| X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | X <sub>6</sub> | X <sub>7</sub> | y  |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----|
| 1              | 0.3            | 1              | 0.2            | 0.9            | 1              | 10             | 5  |
| 12             | 2              | 3              | 0.3            | 11.8           | 5.9            | 2              | 30 |
| 5              | 1.5            | 5              | 0.1            | 5.9            | 10.6           | 5              | 18 |
| 17             | 1              | 7              | 0.25           | 16.5           | 13.9           | 25             | 40 |

X: Features, Y: Label. If you choose to apply feature reduction schemes on the above table, what are the features you consider, and what are the features you drop? Justify your answer

4. How an outlier/ noise affects the performance of the KNN algorithm? how it can be addressed?
5. If  $R_1=[1, 2, 5, 6, 7]$ ,  $R_2=[7, 3, 6, 4, 2]$ ,  $R_3=[9, 2, 5, 8, 4]$ ,  $R_4=[8, 6, 1, 4, 5]$ ,  
 $P_1=[1, 0, 2, 1, 3]$ ,  $P_2=[0, 1, 4, 2, 5]$ ,  $P_3=[0, 3, 7, 1, 4]$ ,  $P_4=[1, 0, 2, 4, 2]$ ,  $P_5=[1, 0, 1, 0, 1]$ ,  
 $R_1 \dots R_4$  corresponds to the Rows of the Data matrix, and  $P_1 \dots P_5$  corresponds to the principal components arranged in the decreasing order of the eigenvalues. Transform the Dataset  $R_1 \dots R_4$  into three-dimensional dataset



### **Assessment No 3, Sub: Data Analysis and Visualization**

1. Explain the working of dbscan algorithm in detail along with the advantages and limitations. Also define what are core points, boundary points and outlier points. 3M
  2. Explain in detail the working of K Means algorithms and its limitations 3M
  3. Explain the working of the Local Outlier Factor Algorithm for Anomaly Detection 4M
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# DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University), GIRINAGAR, PUNE – 411 025

## END SEMESTER EXAMINATION: AUTUMN 2023

Course Name: Data Analysis and Visualization

Course Code: AM 610

Duration: 03 Hrs

Max Marks: 50

Note:

1. Before attempting, please ensure that you have the correct question paper.
2. Marks are indicated against each question. Calculators are only allowed.
3. Write clearly and legibly. All diagrams to be neatly labeled wherever applicable.
4. Answer all questions.

Q1)

[3+4+3]

a) Explain different types of Data with examples. If some of the data is missing in the dataset, what are the techniques you employ to deal with the missing data?

b) Explain the different scales of measurement with one example for each scale of measurement

c) Compute the mean square error and the mean absolute error for the below data, what is the advantage and disadvantages of these metrics

|                 |     |   |   |   |    |      |    |
|-----------------|-----|---|---|---|----|------|----|
| True Value      | 6   | 3 | 7 | 5 | 9  | 12   | 15 |
| Predicted Value | 7.5 | 2 | 9 | 4 | 10 | 10.5 | 14 |

Q2)

[3+3+4]

a) Explain in detail how Skewness and Kurtosis are computed. What it conveys and its significance.

b) For the data  $A = \{12, 23, 10, 12, 30, 25, 60, 45, 30, 85\}$ , Give out the inferences about the data with respect to Skewness and Kurtosis with appropriate terminologies.

c) Draw the box plot for Data A (refer Q 2(b)).

[3+3+4]

Q3)

- a) Explain the similarities and differences between covariance, correlation and causation, and how they can be used for feature selection
- b) Explain the scenario where there is a need for increasing the dimensionality of the data
- c) Mention any four metrics that can be used for computing the similarity between the data points

[4+2+4]

Q4)

- a) Explain in detail about Local Outlier Factor algorithm with an explanation of mathematical expressions
- b) How the presence of an outlier will affect the performance of the KNN algorithm, K Means algorithm, DBSCAN algorithm with a clear explanation
- c)  $A = \{[10, 21], [13, 19], [6, 35], [14, 1], [25, 5], [33, 10], [25, 2]\}$ . Consider the  $[14, 1]$ ,  $[10, 21]$  as the initial cluster centers for K Means clustering algorithm with number of clusters being 2. Give the cluster centers along with the data points that belong to these cluster centers

[5+5]

Q5)

- a) Explain in detail about the Linear Discriminant Analysis (LDA) with mathematical expressions
- b) Explain in detail the working of Principal Components Analysis with mathematical expressions

-----END OF THE QUESTION PAPER-----



**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
(Deemed University)

**M.Tech / Ph.D Assessment-I (15<sup>th</sup> Sept. 2023)**

|                                                            |                               |
|------------------------------------------------------------|-------------------------------|
| Course (Subject) : <b>Advanced Optimization Techniques</b> | Course Code : <b>AM-603 /</b> |
| <b>Computer Oriented Optimization Techniques</b>           | <b>AM-603D</b>                |
| Duration : <b>01 hr</b>                                    | Max Marks : <b>10</b>         |

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**Note.** Attempt **all** questions from the following

1.   a). Write the Phases of O.R. Model.  
      b). Define Basic Feasible solution  
      c). Discuss Standard form of LPP.
  
2. .Use Graphical method to solve the following L.P.P

$$\text{Minimize } Z = 2x_1 + x_2$$

$$\text{S.to.} \quad 3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
(Deemed University)  
**M.Tech Ph.D MID-II SEMESTER EXAMINATION: 06<sup>th</sup> November, 2023.**

Course (Subject) : Advanced Optimization Techniques

Course Code : AM-603

Duration : 01 hr

Max Marks : 10

**Note:** Attempt All two questions from the following. All questions carry equal marks.

1). A travelling salesman has to visit 5 cities. He wishes to start from a particular city, visit each city once and then return to his starting point. Cost of going from one city to another is shown below. You are required to find the least cost route.

| From City | To City |    |    |    |    |
|-----------|---------|----|----|----|----|
|           | A       | B  | C  | D  | E  |
| A         | -       | 4  | 10 | 14 | 2  |
| B         | 12      | -  | 6  | 10 | 4  |
| C         | 16      | 14 | -  | 8  | 14 |
| D         | 24      | 8  | 12 | -  | 10 |
| E         | 2       | 6  | 4  | 16 | -  |

2). A small project is composed of nine activities whose time estimates are listed in the table as follows :

| Activity                    |       | 1-2 | 1-6 | 2-3 | 2-4 | 3-5 | 4-5 | 5-6 | 6-7 | 7-8 |
|-----------------------------|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Estimated duration ( weeks) | $t_o$ | 3   | 2   | 6   | 2   | 5   | 3   | 1   | 3   | 4   |
|                             | $t_m$ | 6   | 5   | 12  | 5   | 11  | 6   | 4   | 9   | 19  |
|                             | $t_p$ | 15  | 14  | 30  | 8   | 17  | 15  | 7   | 27  | 28  |

i). Find the critical path of the project and project duration.

ii) What is the probability that the project will be completed

- at least 4 weeks earlier than expected.
- Not more than 4 weeks later than expected
- In 31 weeks..

**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
(Deemed to be University)

**M.Tech /Ph.D END SEMESTER RE-EXAMINATIONS : Dec- 2023**

Course (Subject) : **Advanced Optimization Techniques / Computer Oriented Optimization Techniques** / Course Code **AM-603 / AM-603D**

Duration : **03 hrs**

Max Marks : **50**

**Note:** 1 Before attempting , please ensure that you have the correct question paper.

2. Attempt **all Five** questions

3. Students may request the invigilator for required Tables (if any).

- 1). Solve the following L.P.P. by Simplex method

(10 M)

Maximize  $Z = -x_1 + 3x_2 - 2x_3$

S.to.  $3x_1 - x_2 + 2x_3 \leq 7$

$-2x_1 + 4x_2 \leq 12$

$-4x_1 + 3x_2 + 8x_3 \leq 10$

$x_1, x_2, x_3 \geq 0$

- 2 a). Define artificial variable.

(2+8 = 10 marks)

- b). Use Two-Phase method to solve the following L.P.P

Minimize  $Z = 2x_1 + x_2$

S.to.  $3x_1 + x_2 = 3$

$4x_1 + 3x_2 \geq 6$

$x_1 + 2x_2 \leq 4$

$x_1, x_2 \geq 0$

3. Using Kuhn-Tucker conditions solve the following NLPP.

[ 10 M]

Max  $Z = 7x_1^2 - 6x_1 + 5x_2^2$

S.to.  $x_1 + 2x_2 \leq 10$

$x_1 - 3x_2 \leq 9$

- 4). Apply Travelling Salesman Algorithm to solve the following problem ( 10 M)

|          |     | (Cities) |          |          |          |          |
|----------|-----|----------|----------|----------|----------|----------|
| (Cities) |     | I        | II       | III      | IV       | V        |
|          | I   | $\infty$ | 4        | 7        | 3        | 4        |
|          | II  | 4        | $\infty$ | 6        | 3        | 4        |
|          | III | 7        | 7        | $\infty$ | 7        | 5        |
|          | IV  | 3        | 3        | 7        | $\infty$ | 7        |
|          | V   | 4        | 4        | 5        | 7        | $\infty$ |



5. a). Use Steepest Decent Method to solve the following problem. (5+5 = 10 marks)

Minimize  $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$  starting from the point  $x_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$

- b). Use Bellman's Principle of optimality to find the shortest path of the following network whose distance matrices are given below:

|   |   |   |   |
|---|---|---|---|
|   | 2 | 3 | 4 |
| 1 | 2 | 4 | 3 |

|   |   |   |   |
|---|---|---|---|
|   | 5 | 6 | 7 |
| 2 | 7 | 4 | 6 |
| 3 | 3 | 2 | 4 |
| 4 | 4 | 1 | 5 |

|   |   |   |
|---|---|---|
|   | 8 | 9 |
| 5 | 1 | 4 |
| 6 | 6 | 3 |
| 7 | 3 | 3 |

|   |    |
|---|----|
|   | 10 |
| 8 | 6  |
| 9 | 3  |

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**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
(Deemed to be University)

**M.Tech. Autumn Semester End Exam (Dec. 2023)**

**Subject : Linear Algebra and Applications**

**Course Code : AM605**

**Duration : 03 Hrs.**

**Max. Marks : 50**

➤ **Calculators are allowed.**

1. a) Define a linear transformation. [2]  
b) Define rank and nullity of a linear transformation. [2]  
c) Let  $G: R^3 \longrightarrow R^3$  be a linear map that rotates a vector  $V$  about the  $Z$  - axis by an angle  $\theta$ . Find the Kernel and Image of the above transformation. [6]
  2. a) Let  $F: R^3 \rightarrow R^3$  be defined by  $F(x,y) = (4x+5y-z, 2x-y+z, x+y+z)$ . Find the matrix representation of  $F$  relative to the basis  $\{(1,4,0), (2,0,0), (1,2,3)\}$ . [5]  
b) Discuss PCA in detail and give its application in Data mining with an example. [5]
  3. a) Discuss all the cases for finding inverse of a given matrix. Find the Moore Penrose inverse for  $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$  using SVD. [10]
  4. a) Discuss Overfitting and Underfitting for system of equations with examples. [2]  
b) Give the algorithm for finding the largest eigen value and the corresponding eigen vector using power method for a given matrix. [4]  
c) Explain in detail as to how SVD can be used to perform PCA. [4]
  5. a) Compute the QR decomposition of the matrix  $\begin{pmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{pmatrix}$ . [6]  
b) Give the Householder QR algorithm for a given matrix  $A$ . [4]
- .....

**Assessment 1 (06 Sept 2023)**

**Course:** Advanced Statistical Techniques (AM604); **Instructor:** Dr. Sushma Kumari, SoCE&MS, DIAT

**Total marks:** 10marks; marks distribution is indicated against each question

**Questions**

- Q1. Fifty-two percent of the students at a certain college are females. Five percent of the students in this college are majoring in computer science. Two percent of the students are women majoring in computer science. If a student is selected at random, find the conditional probability that **(3 marks)**
- (a) this student is female, given that the student is majoring in computer science;
  - (b) this student is majoring in computer science, given that the student is female.
- Q2. Each of 2 cabinets identical in appearance has 2 drawers. Cabinet A contains a silver coin in each drawer, and cabinet B contains a silver coin in one of its drawers and a gold coin in the other. A cabinet is randomly selected, one of its drawers is opened, and a silver coin is found. What is the probability that there is a silver coin in the other drawer? **(3 marks)**

**Attempt any two from the following**

- Q3. A pair of fair dice is rolled. Let E denote the event that the sum of the dice is equal to 7.
- (a) Show that E is independent of the event that the first die lands on 4. **(1 mark)**
  - (b) Show that E is independent of the event that the second die lands on 3. **(1 mark)**
- Q4. Suppose that n independent trials, each of which results in any of the outcomes 0, 1, or 2, with respective probabilities .3, .5, and .2, are performed. Find the probability that both outcome 1 and outcome 2 occur at least once. **(2 marks)**
- Q5. What is conditional independence of A and B given C? Show with examples that the conditional independence of A and B neither implies, nor is implied by, the independence of A and B. **(2 marks)**



**Assessment 2 (13 Sept 2023)**

**Course:** Advanced Statistical Techniques (AM604)

**Instructor:** Dr. Sushma Kumari, SoCE&MS, DIAT

**Total marks:** 10marks; marks distribution is indicated against each question

**Questions**

Q1. Let  $X$  and  $Y$  be two jointly continuous random variables with joint pdf

$$f(x, y) = \begin{cases} 2, & y + x \leq 1, x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Find the covariance and correlation of  $X$  and  $Y$ . (6 marks)

**Attempt any two from the following**

Q2. If  $D$  is an indicator random variables for the event  $A$ , that is

$D = 1$  if  $A$  occurs and  $D = 0$  if  $A$  does not occur.

(c) What is the expectation of  $D$ ? (1 mark)

(d) What is the variance of  $D$ ? (1 mark)

Q3. State and prove the Markov's inequality. (2 marks)

Q4. The size of a first-year class at DIAT is 150 students. From the past data, on an average of only 30% of those accepted for admission will actually attend using a policy of approving the applications of 450 students. Find the probability that more than 150 first-year students attend DIAT. State every assumptions and reasoning in solving this problem. (2 marks)

### **Assessment 3 (10 Nov 2023)**

**Course:** Advanced Statistical Techniques (AM604)

**Instructor:** Dr. Sushma Kumari, SoCE&MS, DIAT

**Total marks:** 10marks; marks distribution is indicated against each question

#### **Questions**

- Q1. What are the four types of random sampling? Give one example each. (2 marks)
- Q2. A machine has produced a certain product in the past having a mean thickness of 0.05 inch. To determine whether the machine is in proper working order a sample of 10 products is chosen for which the mean thickness is 0.053 inch and standard deviation is 0.003 inch.
- (a) State and test the hypothesis that the machine is in proper working order using a level of significance of (i) 0.05 (ii) 0.01. (2 marks)
- (b) State your inference regarding population for both cases of (a). (1 mark)
- Q3. Describe a decision rule to test the hypothesis that a coin is fair if a sample of 64 tosses of the coin is taken and the level of significance of 0.05. (2 marks)
- Q4. Answer the following:
- (a) A sample of five measurements of the diameter of a sphere were recorded by a scientist as 1,2,1,1,2 cm. Determine the unbiased and efficient estimates of (a) the true mean (b) the true variance (1 mark)
- (b) Find the 95% confidence interval for estimating the mean diameter in the above problem (a) . (1 mark)
- (c) What is the degrees of freedom for contingency table of size 4 x 5. (1 mark)

**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY (DU)**

**M.Tech. / Ph.D. EXAMINATION**

**END-SEMESTER: Aug-Dec 2023 (Autumn)**

Course : Advanced Statistical Techniques  
Duration : 3:00 hrs

Course Code : AM-604  
Max Marks : 50 marks

- Note: 1. Marks are indicated against each question.  
2. Attempt all the 05 questions from the following.

- Q1. a) An airplane is built to be able to fly on one engine. If the plane's four engines operate independently, and each has a 1% chance of failing in any given four-hour flight, what is the chance the plane will fail to complete a four-hour flight to its Pune due to engine failure? (4 marks)  
b) Given  $f(x) = k 2^{-x}$  is a probability distribution for a random variable that can take on the values  $x = 0, 1, 2, 3$  and 4. Find  $k$  and the cumulative probability distribution  $F(x)$ . (4 marks)  
c) What are the conditions under which Binomial Distribution tends to Poisson Distribution? (2 marks)
- Q2. Based on the below data, answer the following questions

|   |   |   |    |   |   |   |   |
|---|---|---|----|---|---|---|---|
| x | 5 | 9 | 12 | 2 | 6 | 8 | 3 |
| y | 7 | 8 | 8  | 4 | 6 | 8 | 5 |

- a) Find the linear correlation coefficient of the data. What does it mean? (4 marks)  
b) Find the regression line of the data set. What is the predicted value of  $y$  for  $x = 5$ . (4 marks)  
c) Write a program in R to calculate the mean, median and standard deviation of  $x$  and  $y$ . (2 marks)
- Q3. a) Let the random variable  $X$  be the number of times a fair coin needs to be tossed till two consecutive heads appear for the first time. What is the expected value of  $X$ ? (2 marks)  
b) Explain the concepts of  $p$ -value, Type-I error and Type-II error in a statistical testing. (4 marks)  
c) Design a statistical test for  $\alpha = 0.05$  and  $\alpha = 0.01$  to check if a coin is fair or biased. The experiment is tossed the coin 100 times and observe 60 heads. What is the  $p$ -value in this case? (4 marks)
- Q4. a) What is Causation? Does Causation imply Correlation? Explain with an example. (2 marks)  
b) Let  $X_1, X_2, \dots, X_n$  be a random sample from a normal distribution with unknown mean  $\mu$  and variance  $\sigma^2$ . Find Maximum Likelihood Estimators (MLEs) of mean  $\mu$  and variance  $\sigma^2$ . (5 marks)  
c) Let  $X(t)$  be a stationary random process with a mean of 3 and an auto correlation function of  $9 + 2e^{-|t|}$ . Find the variance of the random process. (3 marks)
- Q5. a) Let  $X$  and  $Y$  be two independent Gaussian random variables, both having mean 0, variance  $1/4$  and  $1/9$ , respectively. Find the probability of  $P(3X \geq 2Y)$ . (3 marks)  
b) A random sample of 395 people surveyed, where each person was asked their interest in riding a bicycle ( $X$ ) and their age ( $Y$ ). The surveyed data is summarized in the following table. Is there is an evidence to conclude that at 0.05 level, the desire of riding a bicycle does not depend on age. (4 marks)

|                      |           | Y (Age)    |            |            |            |
|----------------------|-----------|------------|------------|------------|------------|
| X (Bicycle interest) | Age-group | (18-24)yrs | (25-34)yrs | (35-49)yrs | (50-64)yrs |
|                      | Yes       | 60         | 54         | 46         | 41         |
|                      | No        | 40         | 44         | 53         | 57         |

- c) A random sample of size 100 is taken from an infinite population having the mean 76 and the variance 256. Estimate the probability that will be between 75 and 78. (3 marks)



Assessment – I - (Autumn 2023)

**Max Marks: 10**

2. All programming questions to be answered in Python 3.x

Page 1 | 1

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY  
(Deemed University)

M.Tech/MS/PhD Examinations  
Assessment – II - (Autumn 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM609

Date: 04 – 10 - 2023

Max Marks: 10

Duration: 01 Hr

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.
2. All programming questions to be answered in Python 3.x

|    |                                                                                                                                                                                                                                                                                                                                                                                              |       |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Q1 | Explain the difference between append and extend in the context of lists in Python with the help of an example.                                                                                                                                                                                                                                                                              | [2.5] |
| Q2 | Write a function in Python that takes a 2D NumPy array (8-bit Grayscale image) of size $m \times n$ and returns a 1D NumPy array as the histogram of the image without using inbuilt histogram or hist function..                                                                                                                                                                            | [2.5] |
| Q3 | The following list C contains temperature readings from a sensor in Fahrenheit. Write a <b>single-line</b> statement in Python using map/reduce/filter/lambda to generate a filtered list F of all temperature values above 66 degrees Celsius.<br>C = [139.2, 266.5, 110.3, 99, 154.5, 261, 86.8]<br>F = _____ # fill with your answer<br>$^{\circ}C = (^{\circ}F - 32) \times \frac{5}{9}$ | [2.5] |
| Q4 | Explain the methods get, keys, values, update, and clear in the context of dictionaries data structure. (Method explanation and syntax with example)                                                                                                                                                                                                                                         | [2.5] |

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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY  
(Deemed University)  
M.Tech/MS/PhD Examinations  
Assessment – III - (Autumn 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM609

Date: 31 – 10 - 2023

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|-------------|---|------|----|---|-------|----|---|--------|----|-------|---|---|--|---|---|--|---|---|--|---|---|--|---|---|--|---|---|--|---|---|--|---|---|--|---|---|-------|
| Q1 | <p>What is the difference between a Series and DataFrame in Pandas? Create the following DataFrame using List of Dictionaries.</p> <table><tr><td></td><td>A</td><td>B</td><td>C</td></tr><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>1</td><td>5</td><td>6</td><td>8</td></tr></table>                                                                                                                                                                                                                                                                                                                                            |             | A        | B           | C | 0    | 1  | 2 | 3     | 1  | 5 | 6      | 8  | [2.5] |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | B           | C        |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
| 0  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2           | 3        |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
| 1  | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6           | 8        |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
| Q2 | <p>Complete the following code to get the desired output.</p> <pre>import pandas as _____<br/>L1 = [{"Aman", 45}, {"Ankit", 56}, [{"_____", 67}]<br/>DF = pd._____(L1, _____=["Name", "Marks"], index=_____)<br/>print(DF)</pre> <p>OUTPUT :</p> <table><tr><td></td><td>Name</td><td>Marks</td></tr><tr><td>1</td><td>Aman</td><td>45</td></tr><tr><td>2</td><td>Ankit</td><td>56</td></tr><tr><td>3</td><td>Sunita</td><td>67</td></tr></table>                                                                                                                                                                                               |             | Name     | Marks       | 1 | Aman | 45 | 2 | Ankit | 56 | 3 | Sunita | 67 | [2.5] |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks       |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
| 1  | Aman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 45          |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
| 2  | Ankit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 56          |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
| 3  | Sunita                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 67          |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
| Q3 | <p>What is the confusion matrix? Compute the precision and recall values for the data below:</p> <table><tr><td></td><td>y_actual</td><td>y_predicted</td></tr><tr><td></td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td></tr><tr><td></td><td>0</td><td>0</td></tr><tr><td></td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>0</td></tr><tr><td></td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td></tr><tr><td></td><td>0</td><td>0</td></tr><tr><td></td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>0</td></tr><tr><td></td><td>1</td><td>0</td></tr><tr><td></td><td>0</td><td>0</td></tr></table> |             | y_actual | y_predicted |   | 1    | 1  |   | 0     | 1  |   | 0      | 0  |       | 1 | 1 |  | 0 | 0 |  | 1 | 1 |  | 0 | 1 |  | 0 | 0 |  | 1 | 1 |  | 0 | 0 |  | 1 | 0 |  | 0 | 0 | [2.5] |
|    | y_actual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | y_predicted |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
|    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0           |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |
| Q4 | <p>What is Linear Regression? Compare Batch Gradient Descent and Stochastic Gradient Descent.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | [2.5]       |          |             |   |      |    |   |       |    |   |        |    |       |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |  |   |   |       |

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**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
**School of Computer Engineering & Mathematical Sciences**  
**M.Tech/PhD End Semester- Autumn 2023 Examinations**

Roll No: 

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 2 | 3 | - | 2 | 7 | - | 1 | 0 |
|---|---|---|---|---|---|---|---|

Date: 06-12-2023

Total Number of Questions: [05]

Course Title: Data Science: Tools and Techniques

Course Code: AM609

Time Allowed: 3.0 Hours

Max Marks: [50]

**Instructions / NOTE**

- i. Attempt All Questions.
- ii. Support your answer with neat freehand sketches/diagrams, wherever appropriate.
- iii. Assume any missing data to suit the case / derivation / answer.

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|
| <div>Q1</div> <div>[10 Marks]</div> | <div>Explain the following terms:</div> <div><div>(a) McCulloch-Pitts Neuron</div><div>[02]</div></div> <div><div>(b) Difference between Precision and Recall</div><div>[02]</div></div> <div><div>(c) Implement XOR using a single perceptron unit with multiple inputs.</div><div>[02]</div></div> <div>For the following 3D array:</div> <div><div>a3 = np.array([ [10, 11, 12], [13, <u>14</u>, 15], [16, 17, 18]],</div><div>[20, 21, 22], [23, 24, 25], [26, 27, 28]],</div><div>[30, 31, 32], [33, 34, 35], [36, 37, 38]],</div><div>[28, 41, 22], [23, 34, 25], [28, 27, 78]], )</div></div> <div><div>d) extract [30, 31, 32] using slicing</div><div>[02]</div></div> <div><div>e) extract the highlighted/underlined values using negative indexing</div><div>[02]</div></div> |   |   |   |   |   |   |   |   |
| <div>Q2</div> <div>[10 Marks]</div> | <div><div>(a) Learn <math>y = \theta_0 + \theta_1 x</math> on the following dataset, using gradient descent with MSE as loss, where initially <math>(\theta_0, \theta_1) = (4, 0)</math> and step-size, <math>\alpha = 0.1</math> for 3 iterations.</div><div><table><tr><td>x</td><td>y</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table></div><div><div>(b) Write a function in python to implement AND gate using a perceptron unit.</div><div>[05]</div></div></div>                                                                                                                                                                                                                                                           | x | y | 1 | 1 | 2 | 2 | 3 | 3 |
| x                                   | y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |
| 1                                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |
| 2                                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |
| 3                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |
| <div>Q3</div> <div>[10 Marks]</div> | <div><div>(a) Write a function in python that takes an argument as a list and prints the shuffled list without using the inbuilt shuffle() function.</div><div>[05]</div></div> <div><div>(b) For the below DataFrame named 'df':</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |

| P_No  | Ptype       | P_Manufacturer | Price | Qty |
|-------|-------------|----------------|-------|-----|
| P1001 | Pencil      | Natraj         | 15    | 20  |
| P1002 | Ball Pen    | Reynolds       | 10    | 50  |
| P1003 | Gel Pen     | Flair          | 20    | 100 |
| P1004 | Sketch Pen  | Doms           | 50    | 35  |
| P1005 | Paint Brush | Doms           | 30    | 15  |
| P1006 | Pencil      | Natraj         | 15    | 30  |
| P1007 | Ball Pen    | Reynolds       | 10    | 10  |
| P1004 | Sketch Pen  | Doms           | 50    | 60  |
| P1005 | Paint Brush | Doms           | 30    | 85  |
| P1006 | Pencil      | Natraj         | 15    | 45  |

[5]

- (i) Display the Average price of each type of Product having quantity more than 30.
- (ii) Insert another column in 'df' as 'Total' computed as product of 'Price' and 'Qty'.

Q4

[10 Marks]

- (a) Explain K Nearest Neighbors algorithm.
- (b) Write a program in python to sort elements of a list L in ascending order without using inbuilt functions.
- Eg L=[5,3,1,0,6,77] then output list should be [0,1,3,5,6,77]

[05]

[05]

Q5

[10 Marks]

- (a) Logistic Regression is a classification algorithm or a regression algorithm. Justify your answer by providing suitable explanation.
- (b)

[05]

[05]

| Class | A | B | C |
|-------|---|---|---|
| X     | 1 | 2 | 1 |
| X     | 2 | 1 | 2 |
| X     | 3 | 2 | 2 |
| X     | 1 | 3 | 3 |
| X     | 1 | 2 | 2 |
| Y     | 2 | 1 | 1 |
| Y     | 3 | 1 | 1 |
| Y     | 2 | 2 | 2 |
| Y     | 3 | 3 | 1 |
| Y     | 2 | 1 | 1 |

You want to build a decision tree that determines whether a certain pattern is of type X or type Y. The decision tree can only use tests that are based on attributes A, B, and C. Each attribute has 3 possible values: 1, 2, 3 (we do not apply any thresholding). Compute the is the information gain of each attribute at the root?